TOPIC 03 · HYPOTHESIS TESTS & Z-TESTS

Hypothesis Testing & z-Tests (Proportions & Known-σ Means)

Null vs. alternative hypotheses, test statistic formulas, pooled proportion $\hat{p}_C$, p-value rules, Type I / II errors, and statistical power.

1. Core Testing Framework & Variable Definitions

SYMBOL	NAME	DEFINITION & MATHEMATICAL ROLE
H_0, H_a	Null & Alternative Hypotheses	H_0 = claim of "no effect / no difference" (always uses =, e.g., $p = p_0$). H_a = research claim (<, >, or ≠). **Always written using parameters** (p, μ), never $\hat{p}$ or $\bar{x}$!
p_0, μ_0	Null Value	Specific hypothesized population parameter value assumed to be true under H_0.
z	Standardized Test Statistic	$z = (\text{Statistic} - \text{Null Parameter}) / (\text{Null Standard Deviation})$. Measures how many SDs the sample result lies from H_0.
p-value	Observed Significance	Probability of getting a statistic as extreme as (or more extreme than) our observed data, **assuming H_0 is true**.
α	Significance Level	Decision threshold (commonly 0.05 or 0.01). Equals $P(\text{Type I Error})$ when H_0 is true.
β	Type II Error Rate	Probability of failing to reject H_0 when H_a is actually true.
$1 - \beta$	Statistical Power	Probability that a test correctly rejects a false H_0 at a specific true alternative value.

2. All z-Test Formulas

1-PROPORTION z-TEST (TESTING H_0: $P = P_0$)

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

Variables: $\hat{p} = X/n$ (sample proportion); p_0 (null value); n (sample size).

Important: Denominator and Large Counts condition ($np_0 \geq 10$, $n(1-p_0) \geq 10$) BOTH use p_0 (not $\hat{p}$!), because we assume H_0 is true.

2-PROPORTION Z-TEST (TESTING H_0: $P_1 - P_2 = 0$)

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c(1 - \hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \quad \text{where} \quad \hat{p}_c = \frac{X_1 + X_2}{n_1 + n_2}$$

Variables: X_1, X_2 = success counts in each sample; n_1, n_2 = sample sizes; $\hat{p}_c$ = pooled (combined) sample proportion under H_0: $p_1 = p_2$.

Large Counts Check: Verify $n_1\hat{p}_c$, $n_1(1-\hat{p}_c)$, $n_2\hat{p}_c$, $n_2(1-\hat{p}_c) \geq 10$.

1-SAMPLE Z-TEST FOR A MEAN (ONLY WHEN POPULATION SD Σ IS KNOWN)

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

Variables: $\bar{x}$ = sample mean; μ_0 = null mean; σ = **known population** SD; n = sample size. (If σ is unknown and you use sample s, you MUST use a t-test!)

3. Computing the p-Value & Making a Decision

ALTERNATIVE HYPOTHESIS (H_A)	TAIL TYPE	P-VALUE FORMULA ($Z \sim N(0, 1)$)
H_a: param > null	Right-tailed	$P(Z \geq z_{\text{obs}})$ = `normalcdf(z, ∞, 0, 1)`
H_a: param < null	Left-tailed	$P(Z \leq z_{\text{obs}})$ = `normalcdf(-∞, z, 0, 1)`
H_a: param ≠ null	Two-tailed	$2 \cdot P(Z \geq \lvert z_{\text{obs}} \rvert) = 2 \cdot$ `normalcdf(|z|, ∞, 0, 1)`

CONCLUSION RULE & TEMPLATE

- **If p-value ≤ α:** "Because the p-value ([val]) ≤ α ([val]), we **reject H_0**. There is convincing statistical evidence that [H_a in context]."
- **If p-value > α:** "Because the p-value ([val]) > α ([val]), we **fail to reject H_0**. We do *not* have convincing evidence that [H_a in context]." (Never say "accept H_0" or "prove H_0 is true"!)

4. Type I Error, Type II Error & Power Matrix

DECISION \ TRUTH	H_0 IS ACTUALLY TRUE	H_0 IS ACTUALLY FALSE (H_A TRUE)
Reject H_0	**Type I Error (False Positive)** Probability = α	**Correct Decision (Power)** Probability = $1 - \beta$
Fail to Reject H_0	**Correct Decision** Probability = $1 - \alpha$	**Type II Error (False Negative)** Probability = β

HOW TO INCREASE STATISTICAL POWER ($1 - B$)

1. **Increase sample size n** (shrinks SE, lowers both β and overlap without raising α). | 2. **Increase α** (e.g., 0.01 → 0.05; trades off higher Type I risk for lower β). | 3. **Larger true effect size $|p_{\text{true}} - p_0|$** | 4. **Reduce variability σ**.